

Pattern Propagation Mechanics in the FAI Network

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Derivation note D1.27 committed that accumulated Full Aspect Integration (FAI) events across a network of CKS-governed Selves constitute a collective evolution mechanism at population scope, in which common governance patterns propagate and specializations preserve. This note, D2.32, formalizes the operational mechanics by which a governance pattern actually moves from one Self to others through the FAI network. The note specifies a six-step propagation pathway — origination, contribution, exposure, absorption decision, integration, and re-contribution — and identifies the governance controls that operate at each step. The note establishes that propagation breadth versus localization is determined by governed decisions at each step, that adaptation during propagation constitutes governed evolution rather than ungoverned mutation, and that specialization preservation is the natural complement of propagation through the same governance mechanism. The note also states the forced-propagation anti-pattern: any architecture that conditions FAI network participation on a receiving Self’s absorption of specific patterns violates the receiving Self’s home governance sovereignty. All claims are in the calibrated-humility register — stating operational requirements and architectural possibilities, not prescriptions or predictions.

1. Position and derivation relationship

This note, D2.32, is the operational decomposition of D1.27. D1.27 committed that accumulated FAI events constitute a collective evolution mechanism at population scope where common patterns propagate and specializations preserve. That commitment named the population-level result; it did not specify the step-by-step operational pathway through which a governance pattern actually moves from one Self to others. D2.32 fills that gap.

The derivation relationship is direct: D1.27 states that propagation occurs through governed FAI events; D2.32 formalizes how — what the chain of governed decisions is, what governance controls operate at which step, and what determines whether a pattern propagates widely or stays localized. D2.32 does not introduce new architectural axioms. Every element of the six-step pathway it describes is inherited from prior commitments: the shared substrate (D2.06), the inspect right (D2.04), the DNA evolution locus (D2.11), the directed selection mechanism, and the contribution permission configurations (D2.31).

The calibrated-humility register applies throughout. This note states what the architecture makes possible and what it operationally requires; it does not prescribe how any given network of Selves will evolve, what patterns will propagate, or at what rates.

2. What a governance pattern is

For the purposes of this note, a governance pattern is any substrate content that a Self's home governance has found effective and has encoded as a stable element of that Self's home DNA. Governance patterns may take several forms: an orchestration rule that reliably coordinates a class of tasks; a schema that structures a domain of coordination knowledge in a way that generalizes well; a content-domain specification that captures how a particular subject area should be represented; a protocol logic for handling a recurring type of coordination challenge. The common property across these forms is that the pattern exists as human-authored, home-governed substrate content — it lives in the home DNA of the originating Self and is available for contribution to shared substrates.

A governance pattern is not a runtime behavior, a model weight, or an instinct-layer property. The instinct/reasoning separation (Paper 2) carries through the FAI boundary without modification: LLM weights and instinct-layer content do not exchange across FAI events. Only DNA-layer and action-layer content are candidates for inter-Self exchange. Governance patterns, as substrate content in the DNA layer, are within the scope of inter-Self propagation; inference behaviors are not.

3. The six-step propagation pathway

A governance pattern — call it Pattern P, originating in Self A — moves through the FAI network by the following pathway. Each step is governed; the pattern advances only when governance at that step authorizes the advance.

Step 1 — Origination. Pattern P exists in Self A's home DNA as an effective orchestration rule, schema, or governance approach. It is home-authored substrate content, present because Self A's governance found it effective and encoded it through directed selection. Pattern P may have originated through Self A's own experience, or may itself have been absorbed in an earlier FAI event — the propagation chain may have been running for several prior hops. At this step, Pattern P is simply substrate content under Self A's governance.

Step 2 — Contribution. Self A contributes aspects containing Pattern P to an FAI event with Self B. Contribution is a governed act: Self A's governance has configured propagation permissions (D2.31) that determine which categories of aspect content may be shared, at what provenance depth,

and under what participation configuration. Pattern P enters the shared substrate only if Self A's governance permits contribution of Pattern P's category. This is the first governed decision point: the pattern cannot enter the inter-Self space without the originating Self's governance authorizing contribution.

Step 3 — Exposure and evaluation. Self B's governance reviews Pattern P in the shared substrate, using the inspect right (D2.04). The inspect right is available at any time during the FAI event; it is architecturally guaranteed, not dependent on Self A's permission. Self B's governance evaluates whether Pattern P is potentially useful in Self B's governance context — whether it addresses a domain Self B operates in, whether its schema structure is compatible, whether its orchestration logic addresses coordination challenges Self B faces. This evaluation is home governance's responsibility: no external authority directs Self B's assessment.

Step 4 — Absorption decision. Self B's governance decides whether to absorb Pattern P via the DNA evolution locus (D2.11). This is a home governance directed selection decision. Self B may absorb, decline, or defer — all three are legitimate governance outcomes. The architecture places no obligation on Self B to absorb. The absorption decision is made by Self B's governance on Self B's terms, independent of any preference by Self A's governance or any other participating Self. This is the second governed decision point, and the most significant: it is the gate through which the pattern either enters Self B's home substrate or does not.

Step 5 — Integration and adaptation. If absorbed, Pattern P enters Self B's home DNA through directed selection. Self B's governance authors the directed selection event — it decides how Pattern P is integrated with Self B's existing substrate content, what modifications are appropriate to Self B's governance context, and what provenance records accompany the absorbed content. The absorbed pattern need not be identical to the original. Self B may adapt Pattern P — refining its scope, adjusting its schema structure, combining it with existing Self B content, or parameterizing it differently for Self B's domain. This adaptation is not ungoverned mutation: it is governed adaptation, authorized by Self B's home governance during the directed selection event, with provenance tracing the absorbed content's origin.

Step 6 — Re-contribution. In a subsequent FAI event, Self B contributes aspects that now contain the adapted version of Pattern P. Self C evaluates it, makes its own absorption decision, may adapt it further, and in a subsequent event may re-contribute it again. The propagation chain continues in this fashion: each hop is a new governed sequence of steps 2 through 6, with the current carrier Self in the role Self A occupied at step 2 and the current receiving Self in the role Self B occupied at steps 3 through 5.

The chain structure is what makes this governed collective evolution. The pattern advances only through a chain of governed authorizations — each hop requires that the contributing Self's governance permit contribution and that the receiving Self's governance authorize absorption. No single governed hop produces population-level propagation; the population-level result is the cumulative product of many such governed hops, none of which was ungoverned.

4. Governance controls at each step

The six-step pathway is governed at every step. The governance controls are not procedural additions layered on top of an otherwise autonomous propagation process; they are constitutive of the propagation pathway itself.

At step 2, the contributing Self's propagation permission configuration (D2.31) controls what enters the shared substrate. The configuration is substrate content authored by the contributing Self's governance; it is the operative control for what the shared substrate receives.

At step 3, the receiving Self's inspect right (D2.04) is the operative mechanism for evaluation. The inspect right is unconditional — it cannot be restricted by the contributing Self's governance — which is what makes evaluation genuinely independent.

At step 4, home governance directed selection authorization (D2.11) is the gate for absorption. The receiving Self's governance decides; the architecture provides no mechanism by which the contributing Self or any other external authority can compel absorption.

At step 5, the directed selection event itself is governed — Self B's governance authors the event, which means it has authority over the adaptation and the integration of Pattern P into Self B's home substrate. Provenance carries through: the absorbed content retains its origin record, at the depth Self B's governance configured for provenance carry-over at its perimeter.

At step 6, the subsequent contribution decision by the now-carrying Self is again governed by that Self's propagation permissions. The adapted pattern that Self B re-contributes is subject to Self B's own contribution configuration — which may differ from the configuration under which Self A originally contributed.

5. What determines propagation breadth versus localization

Whether Pattern P propagates widely across the network or stays localized to a small number of Selves depends entirely on the governed decisions described in §3 and §4. There is no architectural mechanism that determines propagation breadth independently of governance.

Wide propagation is possible when: the originating Self's propagation permissions include Pattern P's category and make it available for broad contribution; many receiving Selves' governance judges Pattern P sufficiently useful to absorb; the receiving Selves' participation configurations allow contribution in Pattern P's domain; and receiving Selves' subsequent re-contribution configurations make the adapted pattern available for further propagation.

Localization occurs when governance at any step declines propagation. If the originating Self's governance restricts contribution of Pattern P, it never enters the inter-Self space. If receiving Selves' governance consistently judges Pattern P inapplicable to their domains, the pattern propagates only as far as the Selves that found it applicable. If a receiving Self absorbs but does not re-contribute, the propagation chain terminates at that Self.

The architecture does not prescribe which of these trajectories will occur for any given pattern in any given network. That is an empirical question, dependent on the governance configurations of participating Selves, the usefulness of the pattern across diverse domains, and the accumulated experience of the network. What the architecture specifies is the mechanism: breadth is a function of governed decisions, not of ungoverned spread.

6. Adaptation during propagation as governed evolution

The pattern that arrives at Self C may have been adapted by Self B during step 5. Self B integrated Pattern P with its own governance context — Self B's existing schema structure, its particular domain, its accumulated orchestration experience. The version Self C evaluates may be a refined schema, a parameterized orchestration rule, a contextualized protocol logic. It is Pattern P's lineage, but not necessarily Pattern P verbatim.

This adaptive propagation is the architecture's mechanism for enabling both common ground and contextual variation simultaneously. Common ground is produced because Pattern P's core structure — the governance insight it encodes — survives through multiple hops. Contextual variation is produced because each absorbing Self's governance adapts the pattern to its own context, and the adapted version is what propagates forward. The result across many hops is not a uniform copy of Pattern P distributed across the network, but a family of related patterns, each adapted to its carrier's context, all tracing provenance back to Pattern P's origin.

Adaptation during propagation is governed at every occurrence. It happens through directed selection events authorized by home governance, with provenance carrying origin information. It is not mutation in the sense of ungoverned change; it is directed selection in the sense of home governance-authorized integration. The distinction matters for the prior-art claim: the architecture specifies governed adaptation as the mechanism, not ungoverned drift.

7. Specialization preservation as complement to propagation

Patterns that are highly specific to a Self's organizational domain may be evaluated and declined by Selves in different domains. A specialized orchestration rule developed for a particular industry's coordination challenges may simply not apply to a Self operating in a different industry. Receiving Selves' governance, exercising the evaluation at step 3, will judge such patterns inapplicable and decline absorption at step 4.

This is specialization preservation as the natural complement of propagation. Specialized patterns are not protected from propagation by any special architectural mechanism; they are preserved as specializations because the governance decisions of other Selves determine them not applicable. The same home governance directed selection mechanism that absorbs widely applicable patterns declines specialized patterns — the mechanism is unified, and what differentiates propagating patterns from preserved specializations is the distributed outcome of many independent governance evaluations.

The result at population scope is a network in which some patterns are widely shared (those that many Selves found applicable and absorbed) and others are uniquely held by the Selves whose domains they fit. This diversity is architecturally produced by the same mechanism — governed evaluation at step 4 — operating across Selves with different governance contexts.

8. Anti-pattern: forced propagation

Forced propagation is the anti-pattern in which a contributing Self's governance attempts to require other participating Selves to absorb specific patterns as a condition of FAI network participation. This anti-pattern violates a foundational commitment of the architecture.

Absorption is always and exclusively the receiving Self's home governance decision. The architecture provides no mechanism by which a contributing Self can compel, condition, or coerce absorption. FAI participation is not contingent on accepting specific patterns into home DNA. A network that imposed such conditions would have replaced distributed home governance with a form of centralized pattern authority — the originating Self or network authority would be determining what enters receiving Selves' home substrates, which is precisely what home governance sovereignty prohibits.

The forced-propagation anti-pattern is a positive prior-art claim about what the FAI network cannot demand. Any architecture that conditions inter-Self coordination on pattern absorption violates the receiving Self's governance sovereignty as this architecture specifies it. The anti-pattern is named explicitly rather than left implicit so that subsequent implementations have a clear prohibition: participation configurations (D2.31) may specify scope, provenance, and exchange terms; they may not specify required absorptions by participating Selves.

9. Operational test

For a governance pattern present in multiple Selves' home substrates, the propagation pathway is operationally testable through the FAI event record and absorption records that the architecture produces and retains.

A system instantiates the governed propagation pathway specified in this note if and only if, for any governance pattern P identifiable in two or more Selves' home substrates, an observer with access to FAI event records and each Self's directed selection records can:

1. Identify a Self in whose home DNA Pattern P (or an ancestral version) was present prior to any FAI event in the chain.
2. Identify the FAI event or events in which that Self contributed aspects containing Pattern P to the shared substrate.
3. Identify the governed contribution permission configuration under which Pattern P entered the shared substrate.

4. Identify each subsequent receiving Self's inspection and evaluation record for Pattern P in the shared substrate.
5. Identify each receiving Self's absorption decision record — including records of Selves that declined absorption, showing that decline was a governance decision and not a system failure.
6. Identify the directed selection events at each absorbing Self, including the adaptation that occurred, the governance authorization of the event, and the provenance record linking the absorbed pattern to its contributed origin.
7. Identify the re-contribution events at each carrying Self, including the propagation permission configuration under which the adapted pattern was made available to subsequent Selves.

A system that cannot support this reconstruction — because it lacks event records, absorption decision records, directed selection records, or provenance carry-over — may allow pattern exchange, but does not instantiate governed propagation as this architecture specifies it. The test distinguishes governed propagation from ungoverned exchange.

10. Conclusion

Pattern propagation in the FAI network is governed propagation. The six-step pathway — origination, contribution, exposure, absorption decision, integration, and re-contribution — has a governed decision point at every step. A governance pattern advances through the network only through a chain of governed authorizations: the originating Self's governance permits contribution; the receiving Self's governance evaluates and decides absorption; home governance-directed selection integrates and adapts the absorbed pattern; the carrying Self's governance authorizes re-contribution. The population-level result — patterns propagating across organizational and domain boundaries, specializations preserved where governance evaluates them inapplicable — is the cumulative product of many such governed hops, none of which was ungoverned.

Adaptation during propagation is the mechanism by which common ground and contextual variation coexist: each absorbing Self's governance integrates the pattern into its own context, producing a family of related patterns rather than identical copies. Specialization preservation is the natural complement: patterns inapplicable to particular domains are declined through the same evaluation mechanism that absorbs applicable ones. Forced propagation — conditioning FAI participation on absorption — is the anti-pattern the architecture explicitly prohibits, preserving each Self's home governance sovereignty over what enters its own substrate.

This is governed collective evolution at population scope, operationally specified.

Source papers

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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